

Topic: SQL Standards

Q. What is SQL? Discuss its features, the importance of SQL Standards, and categorize the different types of SQL commands with examples.

 Definition to Remember

SQL (Structured Query Language) is the standard **declarative language** for communicating with Relational Database Management Systems (RDBMS). It allows users to create, manipulate, manage, and query databases. SQL is governed by **ANSI** and **ISO** standards to ensure cross-platform compatibility.

1. SQL Standards

Standardizing bodies **ANSI** and **ISO** define the SQL standard to ensure compatibility across different database vendors (Oracle, MySQL, SQL Server).

Standard	Key Feature
SQL-86	First official standard
SQL-92	Introduced Joins and expanded query capabilities
SQL:1999 (SQL3)	Introduced object-relational features, Recursive queries, Triggers

Importance: Ensures core SQL commands work consistently across databases, **preventing vendor lock-in**.

2. Features of SQL

- **Declarative Language:** Users specify *what* data they want, not *how* to retrieve it.
- **Easy to Learn:** Syntax closely resembles natural English.
- **Highly Scalable:** Handles single-row queries to multi-million row analytics.
- **Comprehensive:** Covers data definition, manipulation, and access control.

3. Types of SQL Commands

Category	Full Form	Purpose	Key Commands
DDL	Data Definition Language	Define/modify database structure	CREATE , ALTER , DROP , TRUNCATE , RENAME
DML	Data Manipulation Language	Insert, update, delete, retrieve data	SELECT , INSERT , UPDATE , DELETE
DCL	Data Control Language	Manage user access & permissions	GRANT , REVOKE
TCL	Transaction Control Language	Manage transactions & data integrity	COMMIT , ROLLBACK , SAVEPOINT

Examples:

```
-- DDL
CREATE TABLE Student (ID INT, Name VARCHAR(50));
-- DML
INSERT INTO Student VALUES (1, 'John');
-- DCL
GRANT SELECT ON Student TO user_admin;
-- TCL
COMMIT;
```

★ Must-Write Points (for 10 marks)

1. SQL = Structured Query Language; standard language for RDBMS communication.
2. Governed by **ANSI** and **ISO** standards (SQL-86, SQL-92, SQL:1999).
3. SQL is **declarative** — specifies *what* to retrieve, not *how*.
4. **DDL**: structure operations (CREATE, ALTER, DROP, TRUNCATE, RENAME).
5. **DML**: data operations (SELECT, INSERT, UPDATE, DELETE).
6. **DCL**: access control (GRANT, REVOKE).
7. **TCL**: transaction management (COMMIT, ROLLBACK, SAVEPOINT).

⚡ Quick Recall

SQL → ANSI/ISO Standards (SQL-86, SQL-92, SQL:1999) → Declarative → DDL (structure) → DML (data) → DCL (access) → TCL (transactions)